Do Now · retrieval
OOP: testing and debugging

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Insert 12 into a BST node 10 with empty right child. Where does it go?

Q2

What is a reference?

Q3

What is a local variable?

Q4

If an object's Balance=20 and Deposit(5) correctly adds five, what is Balance?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Inspect class state — The object has a Balance property initially set by its constructor.

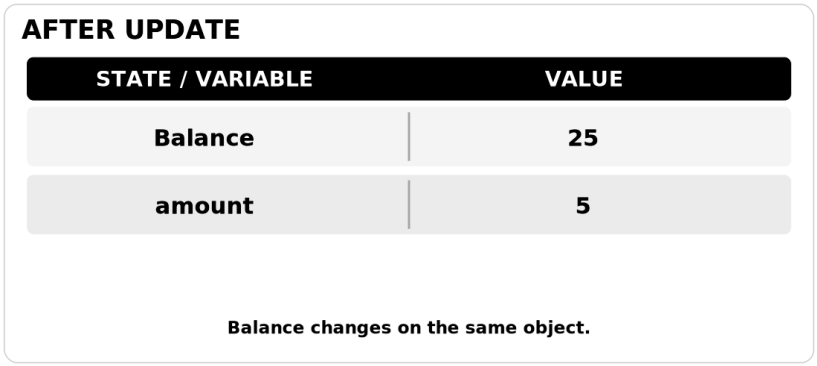
Core idea 2

Trace method inputs — Deposit(5) passes amount=5. Since amount>0, Balance+=5 executes.

Core idea 3

Debug the outcome — Expected Balance=25. If unchanged, check whether the code assigns to Balance or merely calculates a temporary value.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

encapsulation — Hide implementation details and control state through methods/properties.

constructor — Initialises a new object.

field / property — Stores or exposes object state.

breakpoint — Pauses execution at a selected code line.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Find and correct the fault when AddPoints(3) fails to update Score=7.

Follow the worked method

1. Predict first — The local newScore becomes 10, but the field Score remains 7.
2. Model the reasoning — Replace the local-only calculation with Score += points;.
3. Model the reasoning — Retest: after AddPoints(3), Score should equal 10.

Source model · final illustrated step

FIX

STATE / VARIABLE	VALUE
Score	10
instruction	Score += points

Now object state is updated.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Count + 1 does not assign a new value and is not a valid standalone statement in C#.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

An object starts at Count=4 and runs Increment() which performs Count += 1. What is Count after three calls?

Question 2

Correct `public void SetAge(int age) { int Age = age; }` where Age is an existing property. Explain why it fails.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A BankAccount.Withdraw(amount) rejects only amounts <=0, but permits overdrafts. Add validation and give boundary tests for Balance=50.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A method may calculate a correct new value without actually updating the object's property.

Use key vocabulary from memory

1. Explain encapsulation in your own words.

2. Explain constructor in your own words.

3. Explain field / property in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.